

KAMYSHLOV, Russian Federation

WMO: 284510

Lat: 56.850N

Lon: 62.717E

Elev: 127

StdP: 99.81

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.4	-28.6	-34.3	0.2	-31.2	-31.4	0.2	-28.4	8.1	-9.2	7.3	-10.0	1.4	270	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.1	19.7	28.2	18.8	26.4	18.2	21.3	27.7	20.3	26.2	19.3	24.5	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	23.8	18.1	13.2	23.1	17.2	12.5	22.2	62.2	27.9	58.8	26.1	55.5	24.7	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.1	6.2	DB	-34.6	33.1	3.5	1.9	-37.1	34.5	-39.2	35.6	-41.2	36.7	-43.8	38.1	(4)
(5)			WB	-34.7	22.8	3.5	1.4	-37.2	23.8	-39.3	24.6	-41.3	25.3	-43.8	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-14.9	-12.9	-4.7	4.6	11.8	17.0	18.4	16.4	10.4	3.5	-6.0	-12.4	(6)	
	DBStd	13.20	7.35	7.23	5.01	5.35	5.19	4.51	3.91	4.14	4.67	4.94	7.46	7.67	(7)	
	HDD10.0	3528	772	640	457	176	43	4	1	2	51	209	481	693	(8)	
	HDD18.3	5841	1030	873	716	413	210	77	49	89	241	461	731	952	(9)	
	CDD10.0	857	0	0	0	13	98	213	262	201	63	7	0	0	(10)	
	CDD18.3	128	0	0	0	0	7	36	52	29	3	0	0	0	(11)	
	CDH23.3	1279	0	0	0	4	113	369	500	256	37	0	0	0	(12)	
	CDH26.7	352	0	0	0	1	21	107	144	73	7	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.2	2.3	2.7	2.9	2.9	2.5	2.1	2.1	2.3	2.6	2.5	2.4	(14)	
(15) Precipitation	PrecAvg	482	23	17	19	25	43	68	79	61	49	39	33	26	(15)	
	PrecMax	621	41	32	46	64	105	138	170	131	92	87	68	48	(16)	
	PrecMin	324	3	2	1	2	5	27	16	12	8	2	4	7	(17)	
	PrecStd	85	11	8	12	19	26	26	41	29	22	22	16	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	3.3	8.3	23.1	28.9	31.7	32.5	31.5	27.8	18.9	8.3	2.9	(19)	
		MCWB	0.6	0.9	3.8	13.1	15.7	20.1	20.3	20.2	16.2	11.6	5.8	1.5	(20)	
	2%	DB	-0.6	1.5	5.4	18.8	26.5	29.7	30.0	28.8	23.5	15.7	5.6	1.2	(21)	
		MCWB	-1.8	-0.3	1.9	10.5	15.1	18.9	20.4	19.8	15.0	10.1	4.1	0.0	(22)	
	5%	DB	-2.5	-0.6	3.5	15.7	24.0	27.4	28.1	26.1	20.3	12.8	4.0	-0.4	(23)	
		MCWB	-3.5	-2.2	0.8	9.0	14.0	18.3	20.0	18.6	14.2	8.7	2.7	-1.4	(24)	
	10%	DB	-4.5	-2.6	2.0	13.0	21.4	25.2	26.2	23.7	17.7	10.4	2.3	-2.3	(25)	
		MCWB	-5.3	-3.8	-0.1	7.5	12.9	17.6	19.1	17.9	13.0	7.5	1.1	-3.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.4	4.3	13.3	18.0	22.1	22.5	22.0	17.8	12.3	6.0	1.6	(27)	
		MCDB	1.9	2.9	7.5	21.5	24.4	28.7	28.3	29.1	23.1	17.1	7.4	2.6	(28)	
	2%	WB	-1.7	-0.3	2.5	11.0	16.1	20.5	21.5	20.6	16.0	10.6	4.3	0.1	(29)	
		MCDB	-0.8	1.3	4.8	17.3	24.0	27.0	27.7	26.7	21.7	14.6	5.5	1.2	(30)	
	5%	WB	-3.5	-1.9	1.2	9.6	14.9	19.3	20.6	19.4	14.6	9.1	2.7	-1.4	(31)	
		MCDB	-2.6	-0.7	3.0	15.1	22.0	25.3	26.7	24.3	19.5	12.4	3.8	-0.6	(32)	
	10%	WB	-5.2	-3.9	0.0	8.0	13.6	18.2	19.7	18.3	13.4	7.6	1.2	-3.2	(33)	
		MCDB	-4.5	-2.6	1.8	12.6	20.3	24.0	25.1	22.8	17.2	10.2	2.2	-2.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	8.3	8.4	9.4	11.4	10.6	10.3	9.6	8.7	6.7	5.6	6.3	(35)	
		MCDBR	6.4	6.7	8.7	13.7	15.3	13.9	13.6	12.9	12.5	11.1	6.0	6.0	(36)	
	5% WB	MCWBR	5.9	5.6	6.4	7.5	7.1	6.0	5.5	5.5	6.2	6.6	5.1	5.4	(37)	
		MCWBR	6.5	6.0	7.8	12.6	13.6	12.2	12.1	11.6	11.3	10.2	5.8	5.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.284	0.315	0.369	0.415	0.413	0.407	0.430	0.441	0.385	0.352	0.310	0.285	(40)	
		taud	2.044	2.034	1.953	2.082	2.210	2.294	2.229	2.171	2.291	2.254	2.132	2.012	(41)	
	Edn at Noon		558	703	756	791	823	835	803	760	752	668	525	435	(42)	
		Edh at Noon	61	94	134	138	130	121	127	126	97	77	56	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.71	1.72	3.26	4.29	5.14	5.66	5.43	4.18	2.66	1.35	0.64	0.43	(44)	
	RadStd		0.06	0.11	0.19	0.42	0.49	0.50	0.44	0.43	0.24	0.23	0.08	0.06	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page